Reg. No. | | | | | | | | | | | | |

BCACACS 201

Second Semester B.C.A. Degree Examination, June/July 2025
(SEP) (2024 – 2025 Batch Onwards)
DATA STRUCTURES

Time : 3 Hours Max. Marks : 80

Note : *Answer **any ten** questions from Part – **A** and **any one full** question from **each** Unit in Part – **B**.*

PART – A **(10×2=20)**

1. a) Define linear data structure. Give an example.

   b) Define sorting. Why it is necessary ?

   c) Consider the linear array LA (1 : 10). Find the number of elements in an array.

   d) What do you mean by dynamic memory allocation ? Write any 2 functions used in 'C' for dynamic memory allocation.

   e) What is two-way list ? Give its diagram.

   f) Write the advantages of linked list over arrays.

   g) Write an algorithm for PUSH operation using arrays.

   h) Define dequeue. What are its types ?

   i) Write an algorithm to find factorial of a number N using recursion.

   j) Define complete binary tree. Give an example.

   k) What is Graph ? Give an example.

   l) Define the terms with respect to graph.

      i) Adjacent node

      ii) Directed edge.

P.T.O.

PART – B

Unit – I

2. a) Write the classification of data structure and explain.

   b) Trace the following numbers using bubble sort for the following data :

   65, 25, 10, 80, 25, 11, 45, 20.

   c) Write an algorithm to inserting an element into a linear array. **(5+5+5)**

3. a) Explain the various algorithmic notation.

   b) Write an algorithm for selection sort.

   c) Explain the memory representation of one dimensional array. **(5+5+5)**

Unit – II

4. a) Explain the memory representation of linked list.

   b) Write an algorithm for binary search.

   c) Write an algorithm to insert an item at the beginning of a linked list. **(6+5+4)**

5. a) What is linked list ? Explain the different types.

   b) What is linear search ? Write the algorithm for linear search.

   c) Write an algorithm to delete a node following a given node of a linked list. **(6+5+4)**

Unit – III

6. a) Write an algorithm to evaluate postfix expression using stack.

   b) Convert the following infix expression into postfix expression using stack status.

   Q : (((A + B) * D) ^ (E – F)).

   c) Write an algorithm to perform insertion operation on circular queue using array. **(5+5+5)**

7. a) Write an algorithm to convert infix to postfix expression using stack.

   b) Evaluate the following postfix expression showing the stack status.

   P : 3, 5, +, 6, 4, – , *, 4, 1, –, 2, ^ , +.

   c) Write an algorithm to perform insertion operation on queue using linked list. **(5+5+5)**

Unit – IV

8. a) Draw the binary search tree for the following data and traverse in all 3 methods

   66, 26, 22, 34, 47, 79, 48, 32, 78.

   b) Explain the method of representing the graphs using sequential method with an example.

   c) Write an algorithm for Depth First Search (DFS) for a graph. **(5+5+5)**

9. a) With an example, explain sequential representation of binary tree.

   b) Draw the binary tree for the expressions given below. Also traverse in preorder, postorder and inorder.

   A * B + (C ^ D/E/F).

   c) What is adjacency matrix and path matrix, explain with an example. **(5+5+5)**
